Measuring What Matters: A Cost-Validity Framework for Configurable-System Sampling

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***Abstract*—Sampling-based performance prediction evaluates strategies on accuracy and effort, where effort is operationalized as the number of measured configurations. We formalize the gap between this count proxy and the actual resource cost of acquisition. We define $TrainCost$ as the sum of measured per-configuration costs, $Ratio$ as the budget agreement between count and measured cost, and a normalized decision objective $J_\lambda = \widetilde{MRE} + \lambda\widetilde{Cost}$ that separates fixed-size, count-budget, and measured-budget claims.**

Across six evidence families—revisiting published configurable-system benchmarks and extending to performance-energy, workload sensitivity, hyperparameter-learning curves, and Linux kernel releases—we evaluate the conditions under which sample size is a valid cost proxy. We find three regimes: validated-transfer (count conclusions survive), warning (top choices survive but Pareto fronts change), and changed-assessment (recommendations themselves change). The transition between regimes is predicted by cost heterogeneity (CV≥ 0.20 or $p_{90}/p_{10} \geq 1.5$), strategy-specific budget drift ($|\widehat{Ratio} - 1| \geq 0.05$), and target-cost asymmetry.

The result is a validity framework: sampling studies should report cost spread, per-strategy $Ratio$, and measured-cost Pareto fronts alongside accuracy. We provide formal definitions, diagnostic thresholds, and a reproducible artifact.

***Index Terms*—configurable systems, performance prediction, sampling, measurement cost, empirical software engineering**

I. Introduction

The formal problem.: Performance prediction for configurable systems is framed as a sampling problem: measure a subset of configurations, train a model, predict the rest [1]–[5]. A study compares strategies on accuracy and *effort*, where effort is operationalized as the number of measured configurations n. This operationalization is a measurement proxy: it replaces the unobserved resource cost of acquiring training labels with the easily countable sample size. Every measurement proxy embeds a construct-validity claim. Here the claim is that sample count n stands in for the acquisition resource a practitioner would actually pay.

Configurable systems give reasons to doubt this proxy. A video-encoding configuration that enables slow compression multiplies measurement time. An archiver with heavy compression draws more energy. A compiler optimization pass prolongs compilation. In all these settings, two samples with the same n can have very different acquisition costs. The proxy is not random—larger samples tend to cost more—but it may not preserve the decisions a study intends to support.

A measurement-validity gap.: Measurement validity asks whether an observable proxy preserves the inferential conclusions that would be drawn from the true construct. Our construct is measured acquisition cost $TrainCost = \sum_{c \in S} cost(c)$: the resource actually consumed to obtain the training labels. Our proxy is sample count n. The proxy is *decision-valid* if replacing n by $TrainCost$ does not change the recommended strategy, budget, or efficient set. We test this definition across six evidence families.

Figure 1 motivates the gap. Five lrzip strategies all use $n = 16$; count-based effort assigns them identical cost. Measured performance cost does not: pilot-cost sampling spends 0.034% of the full measurement budget, diversity spends 0.567%, a $16.5\times$ gap for nearly identical prediction error (51.3% vs. 52.4% MRE). Two strategies with essentially equal error separate by more than an order of magnitude in real cost.

Figure 2 shows an even stronger failure: a larger sample ($n = 256$) can be cheaper and more accurate than a smaller one ($n = 128$) under measured cost, contradicting the monotonicity intuition that count encodes.

Three-part framework.: We introduce a cost-validity framework with three components. First, a *cost model* defining $TrainCost$, $Ratio$, and min-max normalization that separates three distinct claim types (fixed-size, count-budget, measured-budget). Second, a *budget calibration* analysis using CV, p_{90}/p_{10}, and $Ratio$ to detect when the proxy diverges. Third, a *decision-impact* analysis that measures whether recommendations and Pareto fronts change when the cost axis switches.

We study six evidence families: a replication of Kaltenecker-style configurable-system traces (ICSE'19), an x264 revisit over 17 inputs (ICPE'20), a performance–energy corpus, a workload-impact corpus with 93 system–workload contexts, an LCBench external-validation corpus with 35 OpenML datasets, and seven Linux kernel releases. We ask:

1) To what extent does sample size reflect measured acquisition cost?
2) Does replacing sample size by measured cost change strategy conclusions?
3) Under which conditions is the count proxy valid?

Contributions.

- A formal cost-validity framework for configurable-system sampling, defining $TrainCost$, $Ratio$, and the three-claim taxonomy (fixed-size, count-budget, measured-budget).

Sampling principle. A strategy selects n configurations from the option space; the measured runtime of those configurations is the acquisition cost, and their energy labels train the predictor.

$$\mathcal{C} \xrightarrow[\text{strategy } s]{n=16} S_{s,n}, \qquad TrainCost = \sum_{c \in S_{s,n}} \text{runtime}(c)$$

Excerpt of selected rows.

Strategy	enc.	comp.	method	lev.	cores	runtime
pilotCost	off	off	none	1	1	10.3k
pilotCost	off	off	none	2	3	10.4k
pilotCost	on	off	none	7	1	13.7k
diversity	on	on	zpaq	9	1	1.66M
diversity	off	on	zpaq	9	1	1.61M
diversity	off	on	lzma	9	4	288k

Rows are real selected configurations from the $n = 16$ samples. enc. and comp. are selected/deselected options; runtime is the cost paid before fitting the energy predictor.

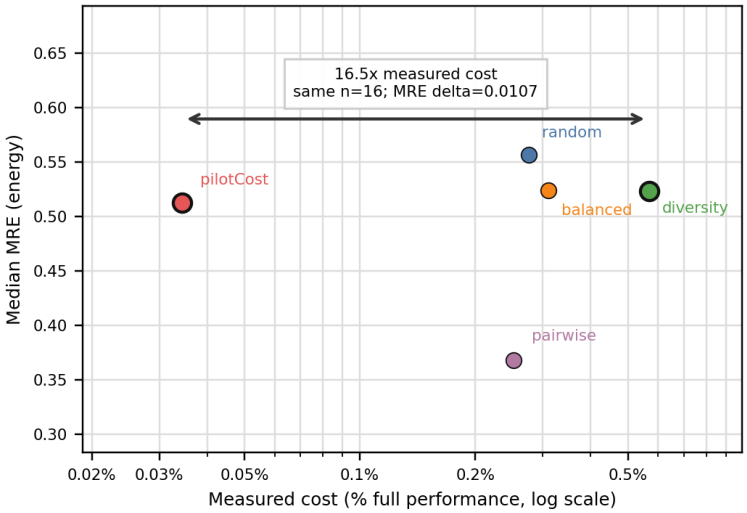


Fig. 1: Integrated running example on lrzip 0.631 compressing a 621 MB UIQ2 workload file (5,184 configurations, 12 option columns, runtime and energy measured over five repetitions). Left: the sampling principle and actual configuration excerpts show how two same-size samples can visit different option regions. Right: all strategies use $n = 16$, but measured performance cost separates nearly equal-error choices: pilot-cost and diversity differ by $16.5\times$ measured cost while MRE changes only from 51.3% to 52.4%.

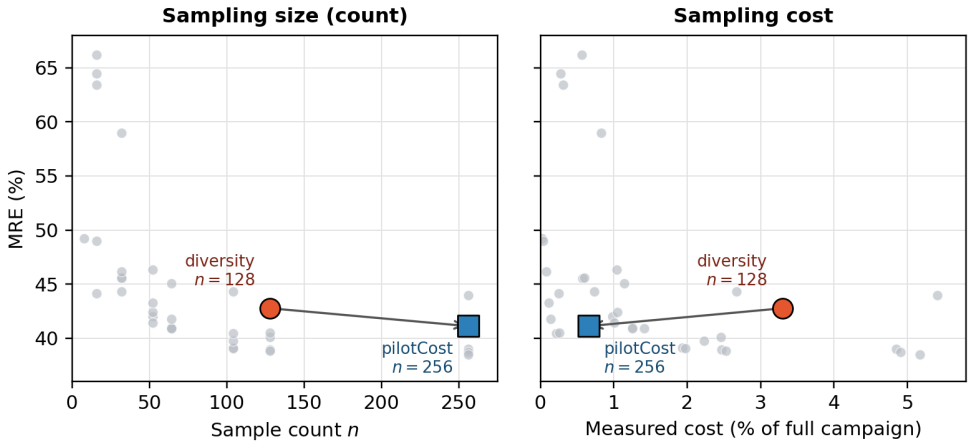


Fig. 2: A larger lrzip sample can be cheaper and more accurate under measured cost. The left panel uses sample count as the cost axis; the right panel uses measured acquisition cost for the same deployable strategy–budget combinations. The orange circle is `diversity` at $n = 128$ (MRE 42.77%, measured cost 3.297% of the full campaign). The blue square is `pilotCost` at $n = 256$ (MRE 41.11%, measured cost 0.658%). Grey points are the remaining deployable combinations in the same scenario.

- An empirical characterization of three validity regimes—validated-transfer, warning, and changed-assessment—across six evidence families.
- Quantitative diagnostic thresholds: CV≥ 0.20, $p_{90}/p_{10} \geq 1.5$, and $|\widetilde{Ratio} - 1| \geq 0.05$ signal when the count proxy needs measured-cost validation.
- A decision audit quantifying that 538/1,666 strategy-budget recommendations change under measured cost, with 57 strict dominance failures.

The remainder of the paper presents the framework (Sections III and IV), its empirical application (Section V), interpretation (Sections VI and VII), positioning (Section VIII), and closes with methodological recommendations (Section IX).

II. Background

A configurable system exposes options whose valid combinations form a configuration space C. For a fixed workload and non-functional property, each configuration $c \in C$ has a measured value y_c such as runtime, output size, or energy. Sampling-based performance prediction trains a regression model $\hat{y}$ from a measured subset $S \subset C$ and evaluates prediction error, typically mean relative error ($MRE = (1/|T|)\sum_{c \in T} |y_c - \hat{y}_c|/|y_c|$; lower is better), on held-out configurations T [3]–[6].

Sampling strategies.: The configurable-systems literature contains several domain-agnostic sampling families. Random sampling draws configurations uniformly when enumeration is available. Solver-based sampling asks SAT/SMT solvers such

as Sat4j or Z3 for valid configurations; randomized solver-based sampling perturbs solver order to increase diversity [7]. t-wise sampling covers combinations of option values [8]–[10]. Distance-based and diversified distance-based sampling select configurations to spread the sample in feature space [4]. When artifacts provide measured pools rather than feature-model synthesis handles, we use pool-based analogues: *balanced* sampling keeps option frequencies close to the pool, *coverage* greedily matches marginal option coverage, *diversity* spreads the sample by Hamming distance, *uncertainty* adds configurations with large predicted residuals, *pairwise* covers option-value pairs, and *pilot-cost* learns a small pilot cost model before choosing configurations predicted inexpensive. Cheap and expensive oracles select the lowest- and highest-cost measured configurations; they are diagnostic bounds, not deployable samplers.

How prior work counts cost.: Prior work has usually equated measurement effort with sample size. Sarkar *et al.* define a composite objective with training cost, testing cost, model cost, and error cost, but then collapse acquisition effort to a term proportional to n under unit per-configuration cost [2]. Westermann *et al.* discuss the number of measurement points needed for a target error [11]; Guo *et al.* vary several measurement budgets (N, $2N$, $3N$, and a larger reference budget M in their notation) [1]; Siegmund *et al.* report Pareto trade-offs between accuracy and number of measurements [3]; Kaltenecker *et al.* compare strategies at common sample sizes [4]. Siegmund *et al.* also report elapsed hours for some heuristics [12], but not as a test of whether n is a valid cost proxy. To our knowledge, the proxy itself has not been evaluated in this literature.

The validity question.: A measurement proxy is valid when it preserves the inferential conclusions that would be drawn from the true construct. Here the proxy is sample count n and the construct is measured acquisition cost $\mathit{TrainCost}$. The proxy is decision-valid if replacing n by $\mathit{TrainCost}$ does not change the recommended strategy, budget, or efficient set. This is the definition we test.

Context sensitivity.: Sampling conclusions already depend on workload and target property. Pereira *et al.* show that x264 strategy rankings change across 17 input videos and two properties [6]. We build on that observation, but ask a different question: not whether prediction accuracy is context-sensitive, but whether the validity of the cost proxy is context-sensitive. The two can come apart. A workload can leave rankings stable while changing the actual measurement budget, or keep the count and measured budgets close while changing the accuracy–cost frontier.

III. Cost-Validity Framework

We present a formal framework for testing whether sample size is a decision-valid proxy for measured acquisition cost.

A. Cost Model

For a run r of strategy s at training size n, let $S_{r,s,n} \subseteq C$ be the selected sample. The usual cost model is the count of measured configurations:

$$\mathit{SizeCost}(s, n) = n. \tag{1}$$

We compare it with measured training acquisition cost:

$$\mathit{TrainCost}(r, s, n) = \sum_{c \in S_{r,s,n}} \mathit{cost}(c). \tag{2}$$

Here $\mathit{cost}(c)$ is the measured resource consumed to acquire the label for configuration c. In most of our study it is wall-clock measurement time; in the performance–energy family it can be energy; in the Linux family it is the uncompressed `vmlinux` size, representing a storage-bound cost. The model does not require the cost dimension to equal the prediction target: one may predict output size while paying time, or predict runtime while paying energy.

To compare systems with different absolute costs, we normalize by the cost of measuring the whole available configuration space:

$$\mathit{FullCost} = \sum_{c \in C} \mathit{cost}(c), \tag{3}$$

$$\mathit{TrainCostPct} = \frac{\mathit{TrainCost}}{\mathit{FullCost}}, \qquad \mathit{SizePct} = \frac{n}{|C|}. \tag{4}$$

The count proxy has good *budget agreement* when the fraction of configurations measured matches the fraction of measured cost spent. We also call this *cost-proxy calibration*: the sample-size proxy is calibrated when a 10% sample spends about 10% of the measured budget. This is not statistical calibration of a prediction model; it is only a check that the cost proxy is on the same scale as the measured resource. We summarize it with a budget ratio:

$$\mathit{Ratio}(r, s, n) = \frac{\mathit{TrainCostPct}(r, s, n)}{\mathit{SizePct}(s, n)}. \tag{5}$$

$\mathit{Ratio} = 1$ means that a 10% sample consumes about 10% of the full measurement cost. $\mathit{Ratio} > 1$ means the strategy selected costlier-than-average configurations; $\mathit{Ratio} < 1$ means it selected cheaper-than-average ones. This is a per-run ratio; one-number summaries report the median ratio unless stated otherwise.

B. Decision Objective and Normalization

For decision-level comparisons, we use a normalized objective that puts MRE and cost on a common scale before combining them. Let $\widetilde{\mathit{MRE}}$ and $\widetilde{\mathit{Cost}}$ denote min-max normalized values:

$$\widetilde{x} = \frac{x - \min(x)}{\max(x) - \min(x)}, \tag{6}$$

computed within each context (system, workload, target) over all strategy–budget combinations. This maps the worst observed value to 1 and the best to 0 while preserving ordering and relative spacing. The combined objective is:

$$J_\lambda^{\text{size}} = \widetilde{\mathit{MRE}} + \lambda \widetilde{\mathit{SizeCost}}, \tag{7}$$

$$J_\lambda^{\text{meas}} = \widetilde{\mathit{MRE}} + \lambda \widetilde{\mathit{TrainCost}}, \tag{8}$$

where the cost weight $\lambda \in \{0.1, 0.25, 0.5, 1, 2, 4, 10\}$ spans accuracy-heavy to cost-heavy choices. Tildes put MRE and cost on the same within-context scale before they are added. A small λ asks an accuracy-heavy question: which combination has the best error unless it is dramatically more expensive? A large λ asks a budget-heavy question: which combination preserves reasonable error while spending little measured cost?

C. Three Claim Types

The framework distinguishes three kinds of claims a sampling study can make, differing in what the cost axis represents:

- **Fixed-size accuracy claim.** Strategies are compared at equal n and evaluated on MRE only. The cost term in J_λ is constant; the ranking collapses to pure accuracy. This claim is always valid because it makes no budget assertion.
- **Count-budget claim.** Strategies are compared at equal n and n is interpreted as the budget. The cost weight λ operates on n directly. This claim assumes n is a valid proxy for the acquisition resource.
- **Measured-budget claim.** Strategies are compared at equal $\mathit{TrainCost}$ and λ operates on the measured resource. This claim requires measuring $\mathit{TrainCost}$ and is the cost-aware gold standard.

The three claims often agree on homogeneous systems, but they diverge when per-configuration costs vary. A study that reports a fixed-size accuracy result but implies a count-budget or measured-budget conclusion has a construct-validity gap.

D. Cost Heterogeneity, Ratio, and Decision Stability

The mathematical relationship between cost spread and decision stability provides the framework's diagnostic predictions. Let CV be the coefficient of variation of per-configuration costs and p_{90}/p_{10} the percentile ratio. When CV is small, $\mathit{TrainCost}$ is approximately proportional to n, so $\mathit{Ratio} \approx 1$ for all strategies and the three claim types converge. When CV is large, strategies that systematically select cheaper or costlier configurations produce $\mathit{Ratio} \neq 1$, and the count-budget and measured-budget rankings diverge.

The divergence is formalized by the difference between J_λ^{size} and J_λ^{meas}:

$$\Delta_\lambda = J_\lambda^{\text{meas}} - J_\lambda^{\text{size}} = \lambda\big(\widetilde{\mathit{TrainCost}} - \widetilde{\mathit{SizeCost}}\big). \quad (9)$$

A strategy–budget combination changes rank when Δ_λ crosses the gap to the next-best combination. Since $\widetilde{\mathit{TrainCost}} - \widetilde{\mathit{SizeCost}}$ depends on Ratio and on the cost distribution, the framework predicts that (a) high cost heterogeneity increases the variance of Ratio across strategies, (b) $|\mathit{Ratio} - 1|$ correlates with the probability of a decision change, and (c) target–cost asymmetry amplifies both effects because the cost and target are independent draws from correlated but distinct distributions.

The framework also counts only training acquisition. Validation measurements are part of the experimental protocol and are the same across strategies; a sampling strategy controls which training configurations it asks us to measure.

IV. Framework Validation: Study Design

The framework makes specific predictions about when the count proxy will fail. We test those predictions across six evidence families.

A. Research Questions

- **RQ1. Cost heterogeneity and budget calibration.** To what extent does sample size reflect measured acquisition cost? We test whether cost spread (CV, p_{90}/p_{10}) and budget agreement (Ratio) reveal proxy divergence.
- **RQ2. Decision impact.** Does replacing sample size by measured cost change the recommended strategy–budget combination or the Pareto-efficient set?
- **RQ3. Validity conditions.** Under which conditions, and with what measurable thresholds, is the count proxy safe?

B. Formal Propositions

The framework yields three testable propositions:

H1: Cost heterogeneity predicts budget disagreement. Systems with CV≥ 0.20 or $p_{90}/p_{10} \geq 1.5$ will show larger strategy-level Ratio dispersion and lower count-measured budget agreement. Systems below these thresholds should have Ratio near 1 across strategies.

H2: Strategy cost bias predicts decision changes. Strategies with median $|\widetilde{\mathit{Ratio}} - 1| \geq 0.05$ will produce a higher rate of top-choice flips and Pareto-front changes. Cost-blind strategies (random, balanced) should have Ratio closer to 1 than strategies that actively explore cost-heterogeneous regions (diversity, uncertainty).

H3: Target–cost asymmetry amplifies the proxy gap. When the prediction target and the acquisition cost are different measured quantities, the decision-change rate will be higher than when they are the same resource, because the strategy that is efficient for the target need not be efficient for the cost.

C. Operational Tests

Cost heterogeneity.: We use the coefficient of variation (CV) because it is unit-free: a CV of 0.20 means the standard deviation is 20% of the mean whether the resource is seconds, joules, or bytes. We pair it with p_{90}/p_{10}, the ratio between the 90th and 10th percentile costs. This second statistic is robust and readable: $p_{90}/p_{10} = 1.5$ means a costly but non-extreme configuration costs 50% more than a cheap but non-extreme one.

Budget calibration.: The ratio Ratio from Equation (5) asks the strict budget question. If a system has 1,000 configurations and a sample measures 100, then $\mathit{SizePct} = 0.10$. If those 100 configurations spend 8% of the full measured campaign, then $\mathit{Ratio} = 0.8$; if they spend 15%, then $\mathit{Ratio} = 1.5$.

Decision impact.: A top-1 flip means the strategy–budget combination recommended by J_λ changes after replacing $\mathit{SizeCost}$ with $\mathit{TrainCost}$. Pareto-front Jaccard overlap measures the broader choice set: Jaccard $= 1$ means the count-cost and measured-cost efficient sets are identical.

D. Evidence Families

We keep six evidence families separate because they stress different validity conditions. Pooling them would give a larger number but a weaker explanation of when the sample-size proxy is reliable.

The REPLICATION-ICSE19 family reuses the published strategy panel—random, solver-based, randomized solver-based, t-wise, distance-based, and diversified distance-based sampling—and changes only the cost axis from sample count to measured performance cost. The REPLICATION-ICPE20 family uses the same logic on x264 over 17 inputs and two properties, testing whether the cost proxy is safer when target and acquisition cost are the same.

The extension families broaden the construct. ENERGY asks whether the issue appears when performance and energy can each be target or cost. WORKLOAD tests whether a conclusion about the cost proxy is a property of the system or of the system–workload pair. LCBENCH moves outside configurable-software performance tables to real runtime learning curves in hyperparameter optimization. LINUX stresses scale: random, balanced, coverage, uncertainty, and diversity sample from large randconfig pools with storage-bound acquisition cost.

Table I gives the factual dataset scale.

E. Protocol

For each tuple of system, workload, target, strategy, training size, and run, we replay or generate the training sample, compute $\mathit{TrainCost}$, train a predictor on configuration options only, and evaluate MRE on held-out configurations. Stochastic strategies use repeated seeds, typically 100 runs. We exclude measured performance, energy, size, and runtime columns from the feature matrix to avoid label leakage.

Artifact.: Our anonymized artifact contains all selected configurations, strategy identifiers, seeds, measured targets, measured acquisition costs, and the full framework implementation: $\mathit{TrainCost}$, Ratio, normalization, Pareto-front computation, and decision-audit scripts.

V. Results

The six evidence families fall into three validity regimes predicted by cost heterogeneity and strategy budget drift. We present the regimes as the headline result, then detail the RQ-specific evidence that supports them.

A. Three Validity Regimes

The *validated-transfer* regime (LLVM, Polly, Dune, HIPAcc, PostgreSQL, x264 time/time) has low cost spread (CV< 0.20 or $p_{90}/p_{10} < 1.5$) and per-strategy Ratio near 1. Count conclusions survive the measured-cost check: the same strategy is recommended under both cost axes.

The *warning* regime (7Z, BerkeleyDBC) has moderate spread but strategy-specific Ratio drift. Top choices mostly survive, but Pareto fronts change. A reader comparing trade-offs would miss measured-cost-efficient alternatives.

The *changed-assessment* regime (lrzip, asymmetric x264 scenarios, heavy-tailed energy systems, many workload contexts, most LCBench datasets, all Linux releases) combines high spread with strategy bias. The recommended strategy–budget combination itself changes.

Figure 3 illustrates four recurring mechanisms that drive regime membership: a same-size cost gap, similar-MRE but different-cost combinations, larger-sample-but-cheaper inversions, and strategy separation under measured cost.

B. RQ1: Cost Heterogeneity and Budget Calibration

Sample size is a useful ordering signal. In every family, larger samples are positively rank-correlated with measured cost. Spearman ρ ranges from 0.710 to 0.996 across families. This confirms that n is not an arbitrary proxy.

The failure is budget calibration. At fixed n, strategies visit very different cost regions. In the running lrzip example, five strategies at $n = 16$ span a $16.5\times$ measured-cost gap with nearly identical MRE. Across the evidence, spread ranges from benign (LLVM $p_{90}/p_{10} = 1.21$) to heavy-tailed (lrzip and BerkeleyDBC $p_{90}/p_{10} \approx 47$, jump3r performance CV 7.104, Linux storage cost CV 0.99–1.43, LCBench runtime CV 0.595–2.247).

Budget ratio confirms the pattern. A nominal 10% sample spends close to 10% of the budget on homogeneous systems, but BerkeleyDBC has median $\mathit{Ratio} = 0.797$, ENERGY reaches $\mathit{Ratio} = 0.168$, workload contexts span 0.661–1.059, and LCBench spans 0.669–1.012. The x264 revisit shows that aggregate calibration is not enough: Henard-style samples tend to be cheaper while t-wise samples tend to be costlier, so a family can pass the coarse "does n order cost?" check and still fail the strategy-specific budget check.

Linux illustrates the split between ordering and calibration. Random-like strategies stay close to the count budget, while diversity costs 1.24–$1.52\times$ its count share at small budgets. An aggregate plot can look safe while a specific strategy is not.

Figure 2 shows the sharpest failure: `pilotCost` at $n = 256$ spends 0.658% of the full campaign while `diversity` at $n = 128$ spends 3.297%; the larger sample is $5.0\times$ cheaper and slightly more accurate.

Answer to RQ1

Cost heterogeneity is widespread. Sample size orders cost reliably, but budget calibration fails when CV≥ 0.20 or $p_{90}/p_{10} \geq 1.5$, and per-strategy Ratio can diverge even when aggregate plots look safe.

C. RQ2: Decision Impact

Replacing sample size by measured cost changes both top recommendations and the efficient choice set. A *recommendation* is the strategy–budget combination that minimizes J_λ for a given weight. A changed recommendation means the study would name a different strategy or budget level. A *choice set* is the Pareto-front of non-dominated combinations; a changed front means some trade-offs visible under one cost model disappear under the other.

TABLE I: Dataset characteristics.

Family	Scope	Configurations	Options/features	NFPs used	Inputs/workloads
REPLICATION-ICSE19	Seven configurable systems [4].	432–68,640 valid configurations.	11–54 options.	One performance-like target per system; same value charged as cost.	One context per system.
REPLICATION-ICPE20	x264 ICPE 2020 [6].	1,152 valid configurations.	16 binary options.	Encoding time and output size, four cost–target scenarios.	17 input videos.
ENERGY	Seven systems from performance–energy corpus.	180–6,840 measured configurations.	System-specific option vectors.	Performance and energy, four scenarios per system.	One workload per system.
WORKLOAD	Nine systems from workload-impact ICSE 2023corpus.	191–6,764 configs; 25,258 total measurements.	System-specific option vectors.	Wall-clock time or throughput.	6–13 workloads per system, 93 contexts.
LCBENCH	35 OpenML datasets LCBench.	2,000 hyperparameter configs per dataset.	7 hyperparameters.	Validation accuracy as target, runtime as cost.	35 dataset contexts.
LINUX	Seven TuxKConfig Linux releases.	19,952–92,562 valid configs per release.	494–497 retained kernel options.	Uncompressed `vmlinux` size as target and cost.	Seven releases, v4.13–v5.08.

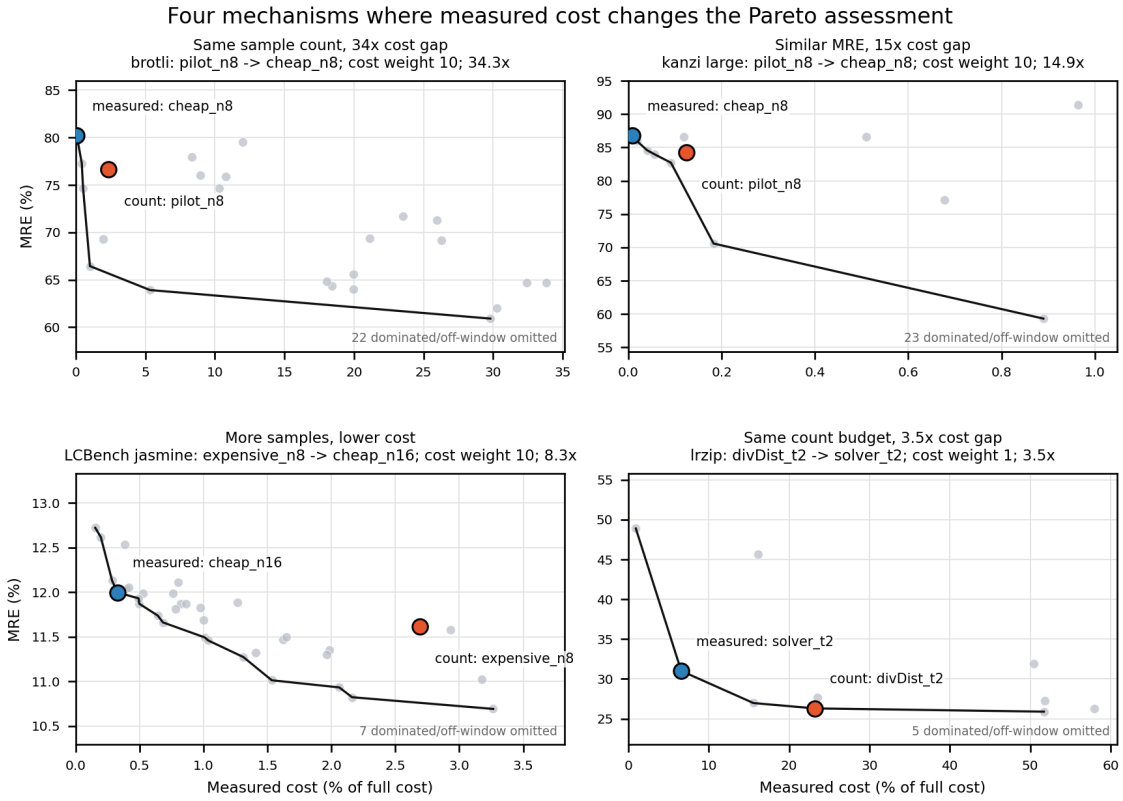


Fig. 3: Mechanism gallery for measured cost. Each panel plots the measured-cost Pareto space for one high-contrast case; lower-left is better. Orange marks the count-cost winner for the stated cost weight, blue marks the measured-cost winner, and the black polyline connects measured-cost Pareto-efficient combinations.

In the REPLICATION-ICSE19 revisit, LLVM, Polly, Dune, and HIPAcc validate the top recommendation, but lrzip changes in $5/7$ cost-weight settings and frontiers change on $5/7$ systems. The x264 revisit is stable in symmetric time/time yet the time-cost/size-target scenario changes $21/119$ input–weight decisions with median Pareto Jaccard 0.375. Beyond the replications, top combinations change in $24/28$ performance–energy scenarios, $72/93$ workload contexts, $33/35$ LCBench datasets, and every Linux release.

A post-hoc decision audit aggregates across primary artifacts. The top strategy–budget combination changes in $538/1{,}666$ context–weight decisions (32%). In $293/538$ changed cases (54%), the count-cost winner is more accurate but costlier—a real trade-off. In $57/538$ changed cases (11%), the count-cost winner is not on the measured-cost Pareto frontier: another combination is no less accurate and no more expensive. Pareto fronts also change broadly: count-cost frontiers miss about 52% of the strategy–budget combinations that are efficient under measured cost.

Answer to RQ2

Measured cost changes the recommended combination in $538/1{,}666$ decisions (32%). The stricter same-budget failure—the count-cost winner is dominated under measured cost—occurs in 57 cases. Pareto fronts change substantially in all high-heterogeneity families.

D. RQ3: Validity Conditions

The six evidence families fall into three regimes predicted by cost heterogeneity and strategy budget drift. The

validated-transfer regime (LLVM, Polly, Dune, HIPAcc, PostgreSQL, x264 time/time) has low cost spread (CV< 0.20 or $p_{90}/p_{10} < 1.5$) and per-strategy $Ratio$ near 1. The *warning* regime (7Z, BerkeleyDBC) has moderate spread but strategy-specific $Ratio$ drift; top choices survive but Pareto fronts change. The *changed-assessment* regime (lrzip, asymmetric x264, heavy-tailed energy systems, many workload contexts, most LCBench datasets, all Linux releases) combines high spread with strategy bias.

Diagnostic thresholds.: Three quantitative signals predict regime membership:

- **CV ≥ 0.20 or $p_{90}/p_{10} \geq 1.5$.** At or above these levels, per-configuration costs are heterogeneous enough for strategy bias to produce budget disagreement. Below them, the count proxy is typically safe.
- **$|\widetilde{Ratio} - 1| \geq 0.05$.** Per-strategy budget drift of 5% or more signals that the strategy systematically selects cheaper/costlier regions. This threshold separates the validated-transfer and warning regimes from changed-assessment.
- **Target–cost asymmetry.** When the acquisition resource differs from the prediction target, the decision-change rate approximately doubles relative to symmetric settings.

These thresholds are empirically derived from the evidence families and serve as practical heuristics rather than universal statistical tests. They correctly classify regime membership for all six families.

Cross-family pattern.: The three regimes recur across all evidence types. Validated-transfer appears in each family's low-heterogeneity contexts (e.g., x264 symmetric time/time, LLVM, low-CV workload contexts). The warning regime appears in systems with moderate spread where strategies keep similar cost profiles. Changed-assessment appears wherever costs are heavy-tailed or target–cost asymmetry is present. The pattern is consistent: the count proxy fails under conditions predicted by the framework, not at random.

Mechanism prevalence.: Figure 3 illustrates the four mechanisms that correspond to the three regimes. Same-size cost gaps are the basic warning signal (warning regime). Similar-MRE/different-cost and larger-sample-but-cheaper inversions indicate changed-assessment. Strategy separation under measured cost is the strongest signal: it shows that even familiar same-budget comparisons change when configurations have heterogeneous acquisition costs.

Answer to RQ3

Sample size is a valid cost proxy under three conditions: low cost spread (CV< 0.20 or $p_{90}/p_{10} < 1.5$), per-strategy $Ratio$ near 1 ($|\widetilde{Ratio} - 1| < 0.05$), and symmetric target–cost dimensions. When these conditions are violated, the proxy enters warning or changed-assessment regimes.

VI. Discussion

The three claim types as a theoretical contribution.: The taxonomy of fixed-size accuracy claims, count-budget claims, and measured-budget claims (Section III-C) is the paper's central theoretical construct. These three claim types correspond to different measurement models. The fixed-size accuracy claim treats sample size as a controlled experimental variable; it makes no cost assertion. The count-budget claim treats n as a resource proxy; it assumes that one configuration unit costs the same as another. The measured-budget claim treats the acquisition resource as the cost; it requires measuring that resource.

The three types are often conflated in practice. A study compares strategies at equal n (fixed-size), reports accuracy differences, and then recommends one strategy as more "efficient" or "cost-effective"—an implicit count-budget or measured-budget claim. Our framework makes these claims explicit and provides the tools to test whether they are empirically warranted. The diagnostic thresholds (CV, p_{90}/p_{10}, $Ratio$) are not arbitrary cutoffs; they operationalize the measurement-validity question.

Validity framework as a methodology contribution.: The framework generalizes beyond our specific systems and strategies. Any empirical study that uses a discrete count (configurations, test cases, benchmarks) as a proxy for a continuous resource (time, energy, money) faces the same construct-validity question. Our framework—define the construct, define the proxy, measure the gap, check whether decisions survive—is a general methodology for testing count-based proxies in experimental software engineering.

The three-regime classification (validated-transfer, warning, changed-assessment) is the framework's empirical output. It shows that the proxy is not universally valid or invalid. Its validity is a measurable property of the data-generation process. Reporting only the regime is more informative than reporting only the accuracy.

Relationship to construct validity.: Construct validity in empirical software engineering asks whether the measured variables adequately represent the theoretical concepts they are intended to capture [13]. The sample-size proxy is a classic construct-validity threat: the observable (count of measurements) is used as the measure of an unobservable (acquisition cost). Our framework formalizes this threat by defining the measurement model, specifying the conditions under which the proxy is adequate, and providing operational tests. This bridges between measurement theory and sampling-benchmark practice.

Generalizing the framework to other cost dimensions.: The framework treats $TrainCost$ generically: it is the sum of per-item costs for whatever resource the study considers the acquisition bottleneck. We demonstrate it on time, energy, and storage cost. The same logic applies to other dimensions: monetary cost of cloud compute, bandwidth, or human annotation time. In each case, the framework asks whether the count of items is decision-equivalent to the sum of their individual costs. The diagnostic tools—CV, percentile ratios, per-strategy $Ratio$—transfer directly. The specific thresholds may shift, but the methodology does not.

VII. Threats to Validity

Cost construct.: We define acquisition cost as the sum of measured per-configuration costs in existing artifacts. This captures the benchmarked resource, not every production overhead such as orchestration, retries, or warm-up. The relative questions we ask—whether same-size samples choose cheaper or costlier configurations—are less sensitive to machine-specific absolute times.

Protocol heterogeneity.: The evidence families use different strategy panels, learners, workloads, and cost dimensions, so we report them separately rather than pooling one effect size. RQ1 is learner-independent; RQ2 and RQ3 depend on MRE and could move with another learner or error metric. We keep original learners for the two replication families and report learner-sensitivity artifacts for stronger learners and newly framed families.

Learner choice.: A better learner can improve MRE and change the winning strategy–budget combination, so it is a threat to any claim about a specific top-ranked sampler. It does not, however, define acquisition cost: RQ1 is computed before fitting any model, from the selected configurations and their measured costs. The Linux HGBT sensitivity run is the key sanity check. It substantially improves accuracy, yet measured cost still changes rankings and Pareto fronts on every release. Better predictors can change which combination wins; they do not make sample size a sufficient cost model.

Framework completeness.: The diagnostic thresholds (CV$\geq$ 0.20, $p_{90}/p_{10} \geq 1.5$, $|\widetilde{Ratio} - 1| \geq 0.05$) are empirically derived from our evidence families. They may shift with different cost distributions or strategy panels. We provide them as practical heuristics, not as universal statistical tests.

External validity and multiple tests.: The systems cover compilers, encoders, databases, archivers, energy measurements, workloads, machine-learning runtime, and Linux releases, but they remain benchmark artifacts. Practitioners should transfer the check, not the exact thresholds: inspect cost spread and per-strategy budget ratios on the target campaign. Because we run many per-scope tests, main claims rely on repeated patterns across families; the supplementary artifact records the per-scope statistics, p-values, and effect sizes.

VIII. Related Work

Sampling configurable systems.: Sampling-based performance prediction has studied random, solver-based, coverage-based, distance-based, progressive, and transfer-learning strategies [1]–[5], [14]. Most papers evaluate effort as sample size. Sarkar *et al.*'s cost-efficient model is the closest configurable-system precedent [2], but its acquisition term is still proportional to n. Our contribution is not a new sampler; it is a validity study of that effort axis. We ask whether the count budget is decision-equivalent to the measured resource.

Surrogate and heterogeneous costs.: Replacing a real resource with an easy-to-count surrogate is a recurring validity concern: when items differ in cost, the count and the resource can diverge. This surrogate-cost threat has been documented in mutation testing, where the number of mutants can mislead relative to execution time [15]; we ask the analogous question for configuration sampling. The mechanism is specific to configurable systems: sample size is usually correlated with measured cost, yet strategy-specific budget agreement and decision-level frontiers can still change when acquisition costs vary widely.

Counting effort versus measuring acquisition cost.: This is the gap we target. Across tuning, transfer, and prediction, "effort," "efficiency," and "speedup" are overwhelmingly expressed as counts—number of measurements, evaluations, or queries—because counts are simple to compare and reproduce [16]–[21]. Counts are sound for ordering effort, but they are a budget only if one count unit stands for a fixed amount of acquisition cost. Our decision tests show that when measured cost diverges from count, the recommended strategy–budget combination or the non-dominated set can change.

Construct validity in empirical SE.: Construct validity asks whether measured variables represent the theoretical concepts they intend to capture [13]. In sampling-based prediction, the construct of "measurement effort" is operationalized as n without testing whether n is a monotone transform of the resource it claims to represent. Our framework provides that test. The closest precedent in the configurable-systems literature is the observation by Guizzo *et al.* that the number of mutants is an incomplete surrogate for execution cost in mutation testing [15]. We extend the same logic to configurable-system sampling and add the decision-audit methodology that measures whether the proxy actually changes conclusions.

Context-sensitive evaluation.: Pereira *et al.* show that x264 sampling rankings change across inputs and target properties [6]. We use these context dimensions to test cost-proxy validity: in some inputs the count proxy is safe, in others it is not.

IX. Conclusion

We present a cost-validity framework for sampling-based performance prediction of configurable systems. The framework defines:

- The measured acquisition cost $TrainCost$, the budget ratio $Ratio$, and a normalized decision objective J_λ that separates three distinct claim types (fixed-size accuracy, count-budget, measured-budget).
- Diagnostic statistics (CV, p_{90}/p_{10}, $Ratio$) that detect when the sample-size proxy diverges from the measured resource.
- A decision-audit methodology that measures whether the proxy changes actual recommendations and Pareto-efficient choice sets.

Empirically, across six evidence families, we find three regimes. In the *validated-transfer* regime (low cost spread, $Ratio$ near 1), count conclusions survive. In the *warning* regime, top choices survive but Pareto fronts change. In the *changed-assessment* regime, the recommendations themselves change. The transition between regimes is predicted by CV$\geq$ 0.20, $p_{90}/p_{10} \geq 1.5$, $|\widetilde{Ratio} - 1| \geq 0.05$, and target–cost asymmetry.

A decision audit over the primary artifacts quantifies the risk: $538/1{,}666$ strategy–budget recommendations (32%) change under measured cost, with 57 strict dominance failures where the count-cost winner is not on the measured-cost Pareto frontier.

The main contribution is the framework itself—the formal definitions, diagnostic thresholds, and decision-audit methodology. The practical implication is that future sampling studies should report cost spread, per-strategy $Ratio$, and measured-cost Pareto fronts alongside accuracy, or explicitly state which of the three claim types they support.

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